

Research Internship Proposal — Mechanics (1)

Page 1

Solid mechanics / materials — experimental + modeling

Quick Info

- Candidate: [Your Name]
- Lab: [Research Lab]
- Duration: 3–6 months
- Theme: Mechanical characterization of materials

1. Research Question

How do processing parameters and microstructure influence elastic/plastic behavior under uniaxial loading, and can a simplified constitutive model fit the observed response?

2. Approach

- Literature review on the selected material system and test standards.
- Design of experiments: sample preparation, tensile tests, repeatability.
- Data processing: stress–strain curves, yield point, hardening parameters.
- Model fitting: choose a basic constitutive model and estimate parameters.

3. Deliverables

- Dataset of tests + plots.
- Parameterized model + fitting notebook/scripts.
- Research report and discussion of limitations.